

Ex.no:9

MINI PROJECT – CREDIT CARD TRANSACTION & FRAUD ANALYTICS

FINANCIAL ANALYTICS

Domain and Problem Statement:

Domain: Financial Data Analytics & Fraud Risk Management

Problem Statement: Banks and card issuers struggle to track transaction volume, customer spending patterns, and fraud exposure across merchant categories and geographies because raw transaction logs remain fragmented and unstructured. This project addresses these challenges by transforming raw card-transaction records into structured business intelligence across Power BI and Tableau to evaluate revenue trends, fraud risk, and customer segments.

Dataset Description:

Dataset Name: Credit Card Transactions Fraud Detection Dataset.

Granularity: Each row represents a single card transaction.

Timeframe Covered: 2019 to 2020.

Key Entities & Attributes:

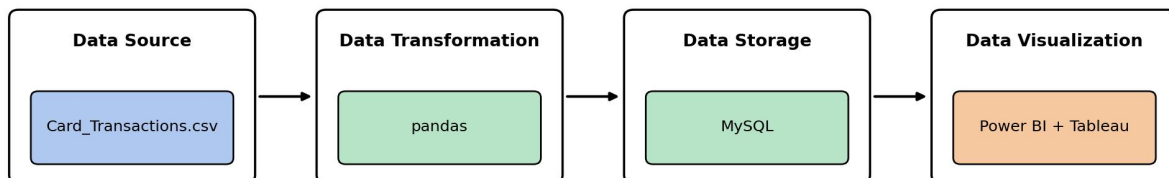
- **Demographics:** customer_id, gender, city, state, job, dob.
- **Transaction Details:** transaction_id, trans_date_trans_time, category, merchant.
- **Financials:** amt (transaction amount), is_fraud, high_value_flag.

Data Source:

Primary Source: Kaggle Open Datasets (Credit Card Transactions Fraud Detection Dataset).

Storage Infrastructure: Cleaned transactional records hosted within a local MySQL database instance (finance_analytics).

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
from sqlalchemy import create_engine, text

# Load dataset
df = pd.read_csv("credit_card_transactions.csv")

# Create a copy for processing
txn = df.copy()

# Remove unnecessary columns
txn.drop(columns=["cc_num", "trans_num", "unix_time"], inplace=True, errors="ignore")

# Handle missing values
txn["job"] = txn["job"].fillna("Unknown")
txn["state"] = txn["state"].fillna("Unknown")

# Remove duplicate records
txn.drop_duplicates(inplace=True)

# Convert date column and derive month
txn["trans_date_trans_time"] = pd.to_datetime(txn["trans_date_trans_time"],
errors="coerce")
txn["transaction_month"] = txn["trans_date_trans_time"].dt.to_period("M").astype(str)

# Derive customer age from date of birth
txn["age"] = (
    pd.to_datetime("today").year - pd.to_datetime(txn["dob"], errors="coerce").dt.year
)

# Flag high value transactions
txn["high_value_flag"] = (txn["amt"] > 500).astype(int)

# Standardize column names
txn.columns = (
    txn.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)

print("Original Shape:", df.shape)
print("Cleared Shape:", txn.shape)
txn.head()
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80
mysql -u root -p
```

MySQL:

```
CREATE DATABASE IF NOT EXISTS finance_analytics;
```

Procedure:

Step 1: Data Preparation & Preprocessing (Python / Pandas)

- Load the raw Kaggle dataset into a Jupyter Notebook / Python script using `pandas.read_csv()`.
- Inspect for null values and handle missing data in job and state columns.
- Feature engineer a high-value transaction flag and a monthly transaction period:

high_value_flag = 1 if amt > 500 else 0

- Filter out invalid rows (e.g., zero or negative amounts) and export the cleaned DataFrame to MySQL using sqlalchemy and `to_sql()`.

 Insert the notebook output here — original vs cleaned row/column counts and the `head()` preview of the `txn` DataFrame.

Step 2: SQL Database Connection & Data Ingestion

1. Open Power BI Desktop → Click Get Data → Select MySQL Database.
2. Enter the Server (localhost:3306) and Database Name (finance_analytics).
3. Enter authentication credentials, select the cleaned flat table (card_transactions / finance_analytics), and click Transform Data to open Power Query.

 Insert the Power BI “Get Data – MySQL database” connection dialog and Navigator preview here.

Step 3: Star Schema Data Modeling (Power BI)

1. Table Splitting in Power Query:

- o Duplicate the imported flat table (card_transactions) three times to create separate entities: Fact_Transactions, Dim_Customer, Dim_Merchant, and Dim_Date.
- o In Dim_Customer, keep gender, state, job, and age. Remove duplicates and add an Index Column as CustomerID.
- o In Dim_Merchant, keep merchant and category. Remove duplicates and add an Index Column as MerchantID.
- o In Fact_Transactions, merge with the dimension tables to map CustomerID and MerchantID, then remove the raw text columns.
- o Click Close & Apply.

2. Establishing Relationships in Model View:

- o Open the Model View tab in Power BI.
- o Connect Dim_Customer[CustomerID] → Fact_Transactions[CustomerID] (1:*).
- o Connect Dim_Merchant[MerchantID] → Fact_Transactions[MerchantID] (1:*).
- o Connect Dim_Date[Date] → Fact_Transactions[trans_date_trans_time] (1:*).
- o Hide the foreign-key columns inside Fact_Transactions to keep the report fields pane organized.

 Insert the Power BI Model View showing Fact_Transactions linked to Dim_Customer, Dim_Merchant, and Dim_Date.

Step 4: Core Calculations & Measures (DAX)

Create a dedicated _Measures table and add the following DAX calculations:

Total Transactions:

Total Transactions = COUNTROWS(Fact_Transactions)

Total Transaction Value:

Total Transaction Value = SUM(Fact_Transactions[amt])

Fraud Rate (%):

Fraud Rate =

```
DIVIDE(
    CALCULATE(COUNTROWS(Fact_Transactions), Fact_Transactions[is_fraud] = 1),
    [Total Transactions],
    0
)
```

Average Transaction Value:

```
Average Transaction Value = AVERAGE(Fact_Transactions[amt])
```

 Insert a screenshot of the Fields pane showing the _Measures table with all four DAX measures listed.

Step 5: Power BI Final Executive Dashboard & Drill-Through

1. Executive Canvas:

- o Place Card Visuals across the top: Total Transaction Value, Total Transactions, Fraud Rate, Average Transaction Value.
- o Add a Line Chart mapped to Dim_Date[trans_date_trans_time] with hierarchy drill-down enabled (Year → Quarter → Month) to show transaction value over time.
- o Add a Clustered Bar Chart for Fraud Rate by Dim_Merchant[category].
- o Add Slicers for category, gender, and state.

 Insert the Executive Canvas: KPI cards, transaction-value trend line, fraud-rate-by-category bar chart, and slicers.

2. Customer Details Drill-Through Page:

- o Create a second page named Customer Details.
- o Drag Dim_Customer[state] into the Drill-through field bucket.
- o Add a detailed Matrix/Table visual showing state, Total Transactions, Total Transaction Value, and Fraud Rate.

 Insert the Customer Details drill-through page, and a screenshot of right-click → Drill through from the main dashboard.

Step 6: Tableau Data Connection & Preparation

1. Open Tableau Desktop / Tableau Public.
2. Under To a Server, select MySQL.
3. Enter Server (localhost), Database (finance_analytics), and user credentials to pull the cleaned dataset onto the canvas.
4. Go to any sheet and build calculated fields:

Total Transaction Value:

```
SUM([Amt])
```

Fraud Rate:

```
SUM(IF [Is Fraud] = 1 THEN 1 ELSE 0 END) / COUNT([card_transactions])
```

Total Transactions:

```
COUNT([card_transactions])
```

Average Transaction Value:

```
AVG([Amt])
```

Select Metric Parameter:

String list containing “Total Transaction Value”, “Total Transactions”, “Fraud Rate”.

Dynamic Measure Formula:

```
CASE [Select Metric]
  WHEN 'Total Transaction Value' THEN [Total Transaction Value]
  WHEN 'Total Transactions' THEN [Total Transactions]
  WHEN 'Fraud Rate' THEN [Fraud Rate]
END
```

 Insert the Tableau data-source connection screen and the Edit Parameter [Select Metric] dialog.

Step 7: Tableau Companion Dashboard & Interactive Story

- 1. Worksheets: Build KPI Cards, Revenue Trend Line Chart (using the Dynamic Measure), Fraud Rate by Category Bar Chart, and a State Map.**
- 2. Global Filters & Actions:**
 - o Drag Category, Gender, and Trans Date Trans Time to Filters → set to Apply to Worksheets → All Using This Data Source.
 - o Drag all sheets onto a new Dashboard tab and click the Use as Filter funnel icon on the State Map visual.
- 3. Drill-down map: clicking a state on the map filters the KPI cards and the Fraud Rate Bar Chart; a “Drilldown to Customer Details” action link opens a dedicated detail sheet.**

 Insert the full Tableau dashboard (KPI sheet, trend line, fraud-rate bar chart, state map) and the map tooltip showing a drill-down action.

Observed Insights:

- **Portfolio Transaction Volume & Value:** The dataset captures 1.85M Total Transactions generating \$131.2M in Total Transaction Value, with an Average Transaction Value of \$70.35.
- **Fraud Exposure:** The overall portfolio reflects a 0.58% Fraud Rate, concentrated heavily in the “shopping_net” and “misc_net” merchant categories.
- **Seasonality Trends:** Transaction volume and value peak during the November–December holiday season, followed by a noticeable dip in February.
- **Geographic & Demographic Trends:** Texas, California, and New York account for the highest transaction volumes, and customers in the 35–50 age band contribute the largest share of high-value transactions.